

Assignment 8: Replicating Interaction Techniques (20P)

Hand in in groups of two.

Estimate about 30 hours of work per group member for the whole assignment.

Goals

You understand how complex interaction techniques can be implemented. You can implement interaction techniques based on their description in a scientific paper and supplementary material.

1 Finding a Suitable Paper 3P

Search for scientific papers that describe the concept and/or implementation of new interaction techniques. **Demonstration** and **short paper** tracks of conferences such as UIST, EICS, MobileHCI, and CHI might be a good starting point. You can also use projects that were discussed in our journal club – either this year or in recent years. Please do not select a paper that has already been covered in the *Replicating Interaction Techniques* assignment in recent years. However, you can use those as an inspiration in terms of scope and complexity. You can find an overview of the papers in the *Replicating Interaction Techniques* GRIPS forum.

First, collect papers you find interesting. Then, exclude everything that is not possible to be implemented within **two weeks**, for example because of hardware requirements, or because you would have to gather a lot of training data first. However, feel free to build upon existing code and/or training data. Then, decide which prototype you want to implement and/or extend. Document this process!

2 Implementation 12P

Implement or extend the interaction technique described in the paper. You do not have to build a feature-complete copy, but the general interaction should be similar. If you decide to extend an existing project, your addition should be focused on interaction. Also, document your approach!

3 Presentation 5P

Present your work at the last ITT session on July 16. There will be a demo session inspired by demo sessions at HCI conferences: Each group will be assigned a desk where they set up their demo. During the demo session, the others can come and try out the demo. You should also be prepared for a short Q&A about the background, interaction technique, and evaluation of the chosen paper as well as implementation process, the problems you faced, and the limitations and advantages.

Additionally, create an entry in the GRIPS forum *Replicating Interaction Techniques* and add a short demo video of your prototype (about 30 to 60 seconds).

Document decision process, implementation, and usage of your prototype. If you are unsure about the project's scope or need access to specific hardware, feel free to ask.